

FinTrust Digital Bank

Data Quality Report — Week 2

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1. Overview

This report presents the data quality assessment of the two datasets provided for the FinTrust project: the customer file and the transaction file. The goal is to confirm that the data is reliable and usable for the analysis in the following parts (SQL, Python, Power BI).

Dataset	Number of rows	Number of columns
FinTrust_Customer_Data	1,500	12
FinTrust_Transaction_Data	12,000	11

Both tables are linked through the Customer_ID column, present in each of them.

2. Missing Values

The customer file (FinTrust_Customer_Data) has no missing values across its 12 columns.

The transaction file (FinTrust_Transaction_Data) has missing values in only two columns:

Column	Missing values	% of total (12,000 rows)
Device_Type	96	0.8%
Location	96	0.8%

96 rows are missing Device_Type, 96 are missing Location, with 4 rows missing both. In total, 188 rows (1.6% of the file) have at least one missing value. All other columns in these rows (amount, status, transaction type, etc.) remain complete.

3. Duplicates

No fully duplicated rows were found in either file. The unique identifiers (Customer_ID and Transaction_ID) also contain no duplicates.

4. Inconsistencies in Categorical Columns

The unique values of each text column were reviewed for spelling, capitalization and stray spaces. No inconsistencies were found: categories are properly standardized in both files.

File	Columns checked	Result
Customer_Data	Gender, Customer_Segment, Account_Type, Preferred_Channel, Account_Status, City, Monthly_Income_Band	No inconsistencies

File	Columns checked	Result
Transaction_Data	Transaction_Type, Channel, Device_Type, International_Transaction, Transaction_Status, Risk_Review_Flag	No inconsistencies

5. Data Types

Numeric columns (Age, Tenure_Months, Digital_Engagement_Score on the customer side; Amount_NGN on the transaction side) are correctly recognized as numbers (right-aligned in Excel). Transaction_DateTime is correctly recognized as a date, with no value stored as text. No data type issues were identified.

6. Unusual Values and Numeric Ranges

Minimum and maximum values for each numeric column were checked to catch inconsistent values (negative age, zero amount, etc.):

Field	Minimum	Maximum	Consistent?
Age	18	65	Yes
Tenure_Months	1	96	Yes
Digital_Engagement_Score	16.7	100	Yes
Amount_NGN	100.17	693,454.45	Yes
Transaction_DateTime	01/01/26	31/03/26	Yes

No suspicious bounds were found: no negative values, no unrealistic ages, no zero-amount transactions.

7. Outliers on Transaction Amounts

The interquartile range (IQR) method was applied to the Amount_NGN column:

Metric	Value
Q1 (1st quartile)	NGN 2,192.22
Q3 (3rd quartile)	NGN 48,484.48
IQR	NGN 46,292.26
Upper bound (Q3 + 1.5×IQR)	NGN 117,922.86
Transactions above the bound	1,474 (12.3% of the file)

These 1,474 transactions are not treated as data entry errors. In a banking context, a high-value transaction can be legitimate (a large transfer, an exceptional withdrawal). They are documented here as values to monitor rather than anomalies to correct and can be cross-checked against the Risk_Review_Flag field during the SQL and Python analysis.

8. Relationship Between the Two Tables

We checked every Customer_ID in the transaction file against the customer file. Result: all 12,000 transactions are linked to an existing customer. No orphaned transactions were found.

9. Cleaning Actions Performed

Only one data quality issue required correction: the missing values in Device_Type and Location.

Column	Action	Rows affected	Justification
Device_Type	Replaced with "Not Provided"	96	Only 0.8% of the file; the other fields in these rows remain usable
Location	Replaced with "Not Provided"	96	Same justification

Full details of these actions are recorded in the Data Cleaning Log (separate Excel file). No rows were removed from the transaction file: the cleaned version still contains all 12,000 rows.

10. Summary and Limitations

Overall, both FinTrust datasets are of good quality: no duplicates, no categorical inconsistencies, correct data types, and an intact customer-transaction relationship. The only point of attention is 188 transactions (1.6%) with missing Device_Type and/or Location, addressed through a standardized replacement.

Limitations and points to keep in mind for the rest of the project:

- The 1,474 high-value transactions (outliers) were not removed; they should be examined further in the SQL/Python analysis, particularly in relation to Risk_Review_Flag.
- Replacing missing values with "Not Provided" introduces a new category in Device_Type and Location: worth keeping in mind if these columns are used for groupings in the Power BI dashboard.
- The data is synthetic and created for educational purposes; Risk_Review_Flag does not represent a real fraud or banking risk decision.